

# DEEP VISUAL FORENSICS PIPELINE

Credential Authentication & Institutional Trust Engine | Squad Hackathon 3.0

## THE PROBLEM

In the modern "Intelligent Economy," digital trust is frequently compromised. Traditional OCR systems are vulnerable to sophisticated digital editing, allowing bad actors to manipulate official seals, signatures, and credentials with pixel-perfect precision. This results in **fraudulent academic certificates, corporate non-compliance, and "ghost" documentation** that bypasses human inspection and basic algorithmic checks.

## THE SOLUTION: TWO-LAYERED "PROOF OF LIFE" ENGINE

Our solution provides a robust forensics pipeline that scrutinizes document integrity at both a textual and pixel level:

- **Textual Layer:** Employs docTR and NLP to detect structural inconsistencies, semantic mismatches, and anomalous kerning/line heights typical of digital tampering.
- **Visual Layer:** Utilizes CNNs for **Error Level Analysis (ELA)** and background reconstruction analysis to identify spliced elements and manipulated seals/logos.

## SQUAD API INTEGRATION

The system leverages the Squad stack to monetize verification and automate institutional incentives:

Squad Checkout API

Squad Payment Modal

Split Payments / Sub-accounts

Corporate entities initiate verification via the **Squad Checkout API**. Upon payment, the system uses **Split Payments** to automatically route royalty percentages to the issuing university's sub-account, ensuring institutional participation while maintaining high operational margins for the platform.

## ALIGNMENT WITH THE FOUR PILLARS

### AI Automation

Automates the forensic workflow of document verification, replacing manual inspection with localized heat maps of manipulation.

### Use of Data

Employs predictive, pixel-level data analysis to secure transactions and protect communities from counterfeit credentials.

### Financial Innovation

Transforms revenue models for issuing institutions through automated royalty sharing via the Squad API.

### **Squad APIs**

Scales a critical trust service using a frictionless payment stack and enterprise-grade fraud prevention.